Chemistry

	Normative 4-year Degree
Major	Chemistry 化學
Degree / Award Title	Bachelor of Science in Chemistry 理學士(化學)
Offering Academic Unit	Department of Chemistry
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2401 English for Science	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	CS1102 Introduction to Computer Studies *Or* CS1302 Introduction to Computer Programming	3
College-specified Courses *(for GREAT Stream)*	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	CS1302 Introduction to Computer Programming	3

College / School Requirements (6-13 credit units) Catalogue Term : Semester A 2025/26

(i) For taking the Majors without Stream (6 credit units)

a. Choose **two** from the following four science courses subject areas:

Biology

Course Code	Course Title	Credit Units
CHEM1200	Discovery in Biology	3

Chemistry

Course Code	Course Title	Credit Units
CHEM1101	Introduction to Chemistry	3
CHEM1300	Principles of General Chemistry	3

Mathematics

Course Code	Course Title	Credit Units
MA1501	Coordinate Geometry	3
MA1502	Algebra	3

Physics

Course Code	Course Title	Credit Units
PHY1101	Introductory Classical Mechanics	3
PHY1201	General Physics I	3

b. **Compulsory attendance** for the following two soft skills courses:

Course Code	Course Title	Credit Units
CSCI1001	Employability for Scientists	0
CSCI1002	Career Lab for Scientists	0

(ii) For taking the Stream of "Global Research Enrichment and Technopreneurship Programme (GREAT)" (13 credit units)

a. Research Methodology

Course Code	Course Title	Credit Units
CSCI2002	Workshop on Research Methodology	1

b. College Requirements I (6 credit units). Choose **two** from the following six courses:

Chemistry

Course Code	Course Title	Credit Units
CHEM1300	Principles of General Chemistry	3
CHEM2006A	Principles of Inorganic Chemistry	3
CHEM2007A	Principles of Organic Chemistry	3

Mathematics

Course Code	Course Title	Credit Units
MA2509	Discrete Mathematics	3

Physics

Course Code	Course Title	Credit Units
PHY1101	Introductory Classical Mechanics	3
PHY1202	General Physics II	3

c. College Requirements II (6 credit units). Choose **two** from the following four courses:

Chemistry

Course Code	Course Title	Credit Units
CHEM2004A	Principles of Analytical Chemistry	3
CHEM2008A	Principles of Physical Chemistry	3

Mathematics

Course Code	Course Title	Credit Units
MA1501	Coordinate Geometry	3

Physics

Course Code	Course Title	Credit Units
PHY1203	General Physics III	3

Major Requirements (55-66 credit units) **Catalogue Term : Semester A 2025/26**

1. Core Courses

(i) For taking the Majors without Stream (38 credit units)

Course Code	Course Title	Credit Units
CHEM2004	Principles of Analytical Chemistry	4
CHEM2006	Principles of Inorganic Chemistry	4
CHEM2007	Principles of Organic Chemistry	4
CHEM2008	Principles of Physical Chemistry	4
CHEM2073	Entrepreneurship Programme In Chemistry 1	3
CHEM3014	Inorganic Chemistry	4
CHEM3015	Organic Chemistry	4
CHEM3016	Physical Chemistry	4
CHEM3027	Analytical Chemistry	4
MA2172	Applied Statistics for Sciences and Engineering	3

(ii) For taking the Stream of "Global Research Enrichment and Technopreneurship Programme (GREAT)" (41 credit units)

Course Code	Course Title	Credit Units
CHEM3014	Inorganic Chemistry	4
CHEM3015	Organic Chemistry	4
CHEM3016	Physical Chemistry	4
CHEM3027	Analytical Chemistry	4
CHEM4086	Independent Research I	8
CHEM4087	Independent Research II	8
MA2172	Applied Statistics for Sciences and Engineering	3
MGT2324	Introduction to Entrepreneurship	3
MGT4305	Developing and Presenting a Business Plan	3

2. Electives

(i) For taking the Majors without Stream (28 credit units)

Course Code	Course Title	Credit Units	Remarks
CS2204	Fundamentals of Internet Applications Development	3	
CS2360	Java Programming	3	
CSCI3001	Grand Challenges in the World	3	
CSCI4002	Industrial Attachment Scheme for Science Students	3	* See footnote
CSCI4005	Overseas Internship Scheme for Science Student	3	* See footnote
CHEM2003	Biochemistry	3	BMS2004 Biochemistry is an equivalent course offered to students from 2017/18.
CHEM2005	Principles of Environmental Chemistry	4	

CHEM3012	Genetics	4	BMS3203 Genetics is an equivalent course offered to students from 2022/23.
CHEM3038	Environmental Sampling and Risk Assessment	4	
CHEM3042	Directed Studies in Biology/Chemistry/Environmental Sciences	4	1-4 credits, * See footnote
CHEM3081	Chemical Biology of DNA and RNA	3	
CHEM3082	Graphene: Fundamentals and Emergent Applications	3	
CHEM3084	Forensic Imaging	3	
CHEM3085	Gemological Science	3	
CHEM4021	Environmental Pollution	4	
CHEM4022	Environmental Toxicology	4	
CHEM4036	Project	6	Students of the GREAT stream cannot take this course. * See footnote
CHEM4037	Seminar Series	3	
CHEM4041	Selected Topics in Chemistry	4	
CHEM4088	Entrepreneurship Programme In Chemistry 2	6	
CHEM2071	Biological Chemistry	4	Courses offered in alternate years.
CHEM3052	Chemistry beyond the Molecule: Supramolecular Chemistry	3	Courses offered in alternate years.
CHEM3053	Computational Chemistry	3	Courses offered in alternate years.
CHEM3055	Green Chemistry	3	Courses offered in alternate years.
CHEM3083	Cosmetic Chemistry	3	Courses offered in alternate years.
CHEM4029	Advanced Analytical Chemistry	4	Courses offered in alternate years.
CHEM4030	Advanced Inorganic Chemistry	4	Courses offered in alternate years.
CHEM4031	Advanced Organic Chemistry	4	Courses offered in alternate years.
CHEM4033	Industrial Chemistry	4	Courses offered in alternate years.
CHEM4034	Environmental Control and Waste Treatment	4	Courses offered in alternate years.
CHEM4035	Environmental Measurements	4	Courses offered in alternate years.
CHEM4043	Food Chemistry	3	Courses offered in alternate years.
CHEM4045	Medicinal Chemistry	3	Courses offered in alternate years.

CHEM4051	Forensic Chemistry	3	Courses offered in alternate years.
CHEM4054	Chemical Bonding and Molecular Spectroscopy	3	Courses offered in alternate years.
CHEM4084	Crystallography/Solid-state Inorganic Chemistry	4	Courses offered in alternate years.
CHEM4085	Testing and Certification Sciences	4	Courses offered in alternate years.
CHEM4089	Techniques and Instrumentation for Chemical Biology	4	Courses offered in alternate years.
CHEM4090	Natural Product Chemistry	3	Courses offered in alternate years.
CHEM4091	Advanced Cosmetic Chemistry	4	Courses offered in alternate years.

*** According to the University Requirement stipulated in the Academic Regulations for Undergraduate Degrees AR4.9.2, normative 4-year degree students admitted in 2022/23 and thereafter are required to choose one of these courses marked with an asterisk. If only CHEM3042 Directed Studies in Biology/ Chemistry/ Environmental Sciences is taken among these courses, it must carry a weighting of 3 or 4 credit units.**

(ii) For taking the Stream of "Global Research Enrichment and Technopreneurship Programme (GREAT)" (14-18 credit units)

Students can select any electives from the current major electives listed above except CHEM4036 Project.

Course Code	Course Title	Credit Units
CSCI4007	Patent Application and Technopreneurship	3

For the Comprehensive Chemistry stream:

Core Course (23-24 credit units)

Course Code	Course Title	Credit Units	Remarks
CHEM1300	Principles of General Chemistry	3	
CHEM4030	Advanced Inorganic Chemistry	4	
CHEM4031	Advanced Organic Chemistry	4	
CHEM4054	Chemical Bonding and Molecular Spectroscopy	3	
CHEM4036	Project	6	To be taken in Year 4*

Choose at least one out of the following two courses:

Course Code	Course Title	Credit Units	Remarks

CHEM3042	Directed Studies in Biology/Chemistry/Environmental Sciences	4	3-4 credits, to be taken in Year 4*
CHEM4037	Seminar Series	3	To be taken in Year 4*

** These courses with independent investigative methodology are required to account for 25% of the final year work load.*

For the Cosmetic Chemistry stream

Electives (14-16 credit units)

Course Code	Course Title	Credit Units	Remarks
CHEM3083	Cosmetic Chemistry	3	
CHEM4091	Advanced Cosmetic Chemistry	4	
CSCI4002	Industrial Attachment Scheme for Science Students	3	Project titles to be agreed by the Hong Kong Society of Cosmetic Chemists (HKSCC)

Choose one out of the following two courses:

Course Code	Course Title	Credit Units	Remarks
CHEM3042	Directed Studies in Biology/Chemistry/Environmental Sciences	4	Project shall be related to Cosmetic Chemistry.
CHEM4036	Project	6	Project shall be related to Cosmetic Chemistry.

For the Forensic Chemistry stream

Electives (18-22 credit units)

Course Code	Course Title	Credit Units
CHEM2003	Biochemistry	3
CHEM2809	Science Versus Crime	3
CHEM3084	Forensic Imaging	3
CHEM4051	Forensic Chemistry	3

Choose at least one out of the following two courses:

Course Code	Course Title	Credit Units
CHEM3017	Molecular Biology	4
CHEM3081	Chemical Biology of DNA and RNA	3

Choose one out of the following four courses:

Course Code	Course Title	Credit Units	Remarks

CHEM3042	Directed Studies in Biology/Chemistry/Environmental Sciences	4	Topics shall be related to Forensic Chemistry
CHEM4036	Project	6	Topics shall be related to Forensic Chemistry
CHEM4037	Seminar Series	3	Topics shall be related to Forensic Chemistry
CHEM4089	Techniques and Instrumentation for Chemical Biology	4	